

Jason Bian

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Education

The University of Texas at Austin
M.S. Computer Science

Expected Dec 2025
GPA: 3.69

University of Michigan, Ann Arbor
B.S.E. Industrial & Operations Engineering

Dec 2019
Major GPA: 3.83

Experience

Amazon.com

2021–Present

Data Platform Engineer – Supply Chain Optimization Technologies

2022–Present

Deep Reinforcement Learning – Inventory Optimization

- Co-developed RL agent and inference pipelines for inventory purchasing decisions with active buying actions against 7% of US retail
- Built and maintained real-time inference services for RL model serving across production purchasing environments
- Defined requirements, support patterns, and long-term planning needs with 20+ software and data science partner teams

Coherent Probabilistic Hierarchical Neural Forecaster

- Designed and scaled inference pipelines for 6 core DL forecasting models with 1102+ weekly runs across all zip, merchant, ASIN, and marketplace hierarchies
- Reduced latency of ~550 forecasting input signals and shortened end-to-end runtime of 4 DL models by 64%
- Scaled 150 daily API calls to 630 over 6 months; reduced backtesting turnaround from 48 hours to 5 hours for inventory transfer decisions
- Provided 24/7 Sev-2 support for streaming and batch inference execution pipelines
- Maintained yearly PR count of ~145 across 24 packages with 2.3 avg revisions

Worldwide Sustainability – Emissions Optimization

- Supported transfer decisions targeting ~1.5% of US total emissions via green shipping option; ~524K ktons of fulfillment network abatement in 2023
- Designed Python and Scala applications for data externalization to external vendors for ship-with-Amazon-scale emission values
- Increased high-specificity carbon data availability for warehouse transfer events from 45% to 77% via attribution models
- Increased total pipeline test coverage from 33% to 90% for inbound and warehouse transfer emission pipelines
- Extended alpha/beta environments, CI/CD, logging, integration testing, and orchestration frameworks against ~35.3B read/write events per day
- Grew internal carbon consulting customer base from 27 to 55 teams

Microsoft

2020–2021

Program Manager – Azure Decision Science

Azure Compute Buffer Sizing

- Developed demand arm of Microsoft's multi-echelon capacity management strategy
- Managed buying plans for Azure data-center compute with ~\$55M of monthly demand sensitivity
- Benchmarked COVID volatility and white-noise decompositions for demand signals

Azure Growth & Offer Restrictions

- Improved system coverage of offer restriction/quota management models from 30% to 45% of Azure services
- Built CVP-level analytics infrastructure for growth and weeks-of-supply in monthly capacity reviews

OptimChain

2020–2022

Founder

- Founded consulting shop for Azure cloud migrations, pre-sale POCs, data estate development, self-service notebook pipelines
- Acquired \$45K YTD revenue from consulting and leasing engagements over 3 months
- Sold supply chain solutions via implement-then-hands-off model for four Midwest customers
- Migrated aging manual systems with 1800 hours of labor saved

- Developed custom ingestion frameworks for Azure migrations using Microsoft-sponsored design patterns
- Managed contract negotiation, RFPs, staffing for ~ 7 projects; led 3 US-based and 4 offshore resources

Technical Skills

Languages: Java, Python, Scala

Frameworks: TypeScript, JavaScript, Shell (Bash), Claude Code, Spark/PySpark, PyTorch, Airflow, Django, Flask, React, NumPy, Pandas

Infrastructure: AWS (EMR, Glue, SageMaker, Lambda, Redshift, SNS, CloudWatch, CDK, S3, ECS, Step Functions, Kinesis), Azure (Data Lake, Data Factory, Event Hub, Synapse, PostgreSQL), Apache (Parquet, Presto, Beam, Flink, Avro, Iceberg, Hazelcast), Databricks Delta Lake, GitHub Actions, Jenkins

ML & Optimization: R, Reinforcement Learning, Deep Learning, Hierarchical Forecasting, ARIMA, Convex Optimization, Multivariate Regression, Linear/Integer Programming, Stochastic Processes, Monte Carlo Simulation, Design of Experiments

Projects

Riot Match History Classifier: Guest lecture for INFO SYS 201 on League of Legends API extraction and decision tree classification

Cloud Hydroponics: Remote flow monitoring solution for automated hydroponic systems

Other

House-van construction, minimalist travel, sound engineering